

QUANTUM CRYPTOGRAPHY

Quantum Computing Explained, by David McMahon, Chapter : 11, page: 239

Bennett and Brassard received 2026's Turing Award, equivalent of Nobel Prize in Computing.

Turing Award Goes to Inventors of Quantum Cryptography



<https://www.nytimes.com/2026/03/18/technology/turing-award-winners-quantum-cryptography.html>

Public Key Cryptography

- In public key cryptography, each user has a pair of keys: a public key and a private key.
- The public key is freely distributed and can be shared with anyone, while the private key is kept secret.
- Messages encrypted with the public key can only be decrypted by the corresponding private key.
- It is computationally infeasible to deduce the private key based on the public key.
- public key cryptography is the RSA algorithm

Example

- Alice wants to send a secure message to Bob.
- Bob generates a key pair (public key and private key) and shares the public key with Alice.
- Alice uses Bob's public key to encrypt her message and sends it to Bob.
- Only Bob, with his corresponding private key, can decrypt and read the message.

Private Key Cryptography

- Also known as symmetric-key cryptography, private key cryptography uses the same key for both encryption and decryption.
- The key must be kept secret and shared only between the parties involved in communication.
- While efficient, the challenge is securely distributing and managing the secret key.
- Advanced Encryption Standard (AES) is a widely used symmetric key algorithm

Example

- Alice and Bob agree on a secret key (K) between them.
- Alice uses key K to encrypt a message and sends it to Bob.
- Bob, who also knows the secret key K , decrypts the message to read its contents.

Classical Cryptography

- Sender, receiver share common key
 - Keys may be the same, or trivial to derive from one another
 - Sometimes called *symmetric cryptography*
- Two basic types
 - Transposition ciphers
 - Substitution ciphers
 - Combinations are called *product ciphers*

Transposition Cipher

- Rearrange letters in plaintext to produce ciphertext
- Example (Rail-Fence Cipher or 2-columnar transposition)
 - Plaintext is HELLO WORLD
 - HE
LL
OW
OR
LD
 - Ciphertext is HLOOL ELWRD

Transposition Cipher

- Generalize to n-columnar transpositions
- Example 3-columnar
 - HEL
LOW
ORL
DXX
 - HLODEORXLWLX

Example

- Arrange so the H and E are adjacent

HE

LL

OW

OR

LD

- Read off across, then down, to get original plaintext

Substitution Ciphers

- Change characters in plaintext to produce ciphertext
- Example (Cæsar cipher)
 - Plaintext is HELLO WORLD
 - Change each letter to the third letter following it (X goes to A, Y to B, Z to C)
 - Key is 3, usually written as letter 'D'
 - Ciphertext is KHOOR ZRUOG

Attacking the Cipher

- Exhaustive search
 - If the key space is small enough, try all possible keys until you find the right one
 - Caesar cipher has 26 possible keys
- Statistical analysis
 - Compare to 1-gram model of English
 - CryptoQuote techniques

Cryptographic keys

- Secure communication can be accomplished through key distribution, a method by which two parties who want to communicate privately share a key that is used to encrypt or scramble a message.
- Currently cryptographic keys are generated using mathematical algorithms that are hard but not impossible to break. In contrast, quantum cryptography is a method of key distribution that relies on the laws of physics to create a key.
- While not 100% safe, quantum cryptography offers huge advantages over traditional methods.

Example

- Before we get into quantum cryptography, let's look at a toy example to see how messages can be encrypted.
- Suppose that two parties, denoted Alice and Bob as usual, want to share a private message.

$A \rightarrow 0000$

$B \rightarrow 0001$

$C \rightarrow 0010$

$D \rightarrow 0011$

- A very simple way to secure the message is to create a key k that we add to each character before it's transmitted.
- if the message is m Alice encrypts it with the key k by creating the transmitted string t in the following way

$$t = m + k$$

Example

- To be specific, suppose that $k = 3 = 0011$. We add this to each character in a given message. So the strings above become

$A \rightarrow 0000 \rightarrow 0011$

$B \rightarrow 0001 \rightarrow 0100$

$C \rightarrow 0010 \rightarrow 0101$

$D \rightarrow 0011 \rightarrow 0110$

- That is, $A \rightarrow D$, $B \rightarrow E$, $C \rightarrow F$, $D \rightarrow G$ for our particular encoding scheme.
- If Alice wants to transmit the word BAD to Bob, she encrypts it using the key and obtains E D G

RSA

- RSA (Rivest-Shamir-Adleman) is a public-key cryptography algorithm.
- It was one of the first practical public-key cryptosystems and is widely used for secure data transmission.
- In RSA, each participant has a pair of keys: a public key that is shared with others, and a private key that is kept secret.
- The public key is used for encryption, while the private key is used for decryption.
- The security of RSA is based on the difficulty of factoring the product of two large prime numbers, which forms the basis of the keys.

A BRIEF OVERVIEW OF RSA ENCRYPTION

- The basic idea behind RSA is to use two keys, one that is public and one that is private. To decrypt a message you've got to have the private key, and the security of the system is based on generating very large numbers that are hard to factor.
- We begin by choosing two prime numbers, which we denote p and q .
- In RSA encryption, very large prime numbers $p > 10^{100}$ are used making factoring very difficult, if not impossible.
- It turns out that factoring products of very large prime numbers appears to be so difficult, in principle (meaning that it doesn't matter how fast or powerful a computer we build), that it would take billions of years to factor them.

RSA Scheme

- Shor's algorithm can factor prime numbers readily on a quantum computer—so other cryptographic methods will have to be used (quantum cryptography) to secure messages.
- Let's see how a simple RSA scheme might work. We begin with two large prime numbers p and q and create their product, which we denote n :

$$n = pq$$

- Now we compute another product that some imaginative number theorists have denoted the totient:

$$\phi(n) = (p - 1)(q - 1)$$

RSA Scheme

- Next, let $1 < e < \phi(n)$ such that the only common factor of e and $\phi(n)$ is 1. We generate the public key using n and e . To create the private key, we use

$$d = \frac{1 \bmod \phi(n)}{e}$$

- The private key consists of d and n . Notice that if we knew d , since we know e (which is available on a public channel), we would be able to decrypt the message.

RSA Scheme

- Returning to RSA encryption, a message m is encrypted in the following way:

$$c = m^e \bmod n$$

- The intended party has the private key d in their possession. They can decrypt the message, since

$$c^d = m^{de} \bmod n$$

- Now it so happens that since $n = pq$,

$$m^{ed} = m \bmod p$$

$$m^{ed} = m \bmod q$$

- Some results from number theory tell us that this means that

$$c^d = m \bmod n$$

Example

Let's see how the RSA system works, in principle, by working an example with some very small numbers. Let $p = 3$, $q = 11$, and show how we would encrypt and decrypt a message $m = 6$ using the RSA procedure outlined above.

First, we create the product

$$n = pq = (3)(11) = 33$$

Next, we create the totient

$$\phi = (3 - 1)(11 - 1) = (2)(10) = 20$$

Let $e > 1$ such that e and $\phi = 20$ have no common factors except 1. The smallest number that satisfies this criterion is $e = 3$. To find d , find the smallest x such that

$$de = 1 + x\phi$$

- where d is a natural number. In our case we have

$$d = \frac{1 + 20x}{3}$$

- The smallest value is $x = 1$, giving

$$d = \frac{1 + 20}{3} = \frac{21}{3} = 7$$

- The encrypted message that is transmitted for $m = 6$ is

$$c = m^e \bmod n = 6^3 \bmod 33 = 18$$

- Alice transmits this signal to Bob, who holds the private key $d = 7$ in his possession. He decrypts the message using

$$m = c^d \bmod n = 18^7 \bmod 33 = 6$$

- The RSA encryption algorithm works for many applications in the present day. However, as we mentioned earlier, Shor's algorithm demonstrates that a workable quantum computer can easily crack an RSA encryption scheme, since $n = pq$ can be readily factored.

Syllabus: BB84, B92, E91 Algorithms

BASIC QUANTUM CRYPTOGRAPHY

- Quantum cryptography is a method that uses quantum mechanics rather than simple numerical algorithms to generate a secret key. This is known as quantum key distribution, or QKD.
- To implement QKD, two communications channels are used between Alice and Bob.
- Ordinary public channel-Encrypted messages are sent over this line. In addition a second piece of the QKD puzzle is used—a quantum communications channel over which the quantum key is distributed. In practice, this is done using individual photons in different polarization states.
- So if Eve taps into the line, she has to make measurements—disturbing the system in such a way that Alice and Bob can detect her presence.

BB84

- The first type of quantum cryptography we will look at is called the BB84 protocol. It's named after its discoverers (Bennett and Brassard) and the year when the protocol was first published. There are three key principles used in BB84 QKD:
 1. The no-cloning theorem—quantum states cannot be copied. As a result Eve cannot tap a quantum communications channel, copy the quantum states used to create the key for herself, and send the originals down the line to Bob.
 2. Measurement leads to state collapse (wave function collapse). A key aspect of QKD is that different bases will be used to create a bit string. When we make measurements in one of the given bases, we will cause state collapse such that measurements in the other basis are completely random. In other words, extracting some information about the state disturbs the state of the system.
 3. Measurements are irreversible.

Alice creates an 8-bit string

$$|0\rangle|1\rangle|+\rangle|0\rangle|0\rangle|-\rangle|+\rangle|-\rangle$$

If Bob randomly measures in the following order using the bases $\{|0\rangle, |1\rangle\}$ $\{|0\rangle, |1\rangle\}$ $\{|+\rangle, |-\rangle\}$ $\{|+\rangle, |-\rangle\}$ $\{|0\rangle, |1\rangle\}$ $\{|+\rangle, |-\rangle\}$, describe the bit string that Alice and Bob keep.

We simply throw out the bits where Alice and Bob used a different basis, numbering the bits from 1 to 8. We have the following table:

Alice Bits	0	1	0	0	0	1	0	1
Alice Basis	$\{ 0\rangle, 1\rangle\}$	$\{ 0\rangle, 1\rangle\}$	$\{ +\rangle, -\rangle\}$	$\{ 0\rangle, 1\rangle\}$	$\{ 0\rangle, 1\rangle\}$	$\{ +\rangle, -\rangle\}$	$\{ +\rangle, -\rangle\}$	$\{ +\rangle, -\rangle\}$
Bob Basis	$\{ 0\rangle, 1\rangle\}$	$\{ 0\rangle, 1\rangle\}$	$\{ +\rangle, -\rangle\}$	$\{ +\rangle, -\rangle\}$	$\{ +\rangle, -\rangle\}$	$\{ 0\rangle, 1\rangle\}$	$\{ +\rangle, -\rangle\}$	$\{ +\rangle, -\rangle\}$
Match	Yes	Yes	Yes	No	No	No	Yes	Yes
Keep	Yes	Yes	Yes	No	No	No	Yes	Yes

The resulting bit string is created by keeping bits in positions 1, 2, 3, 7, 8 and discarding the bits in positions 4, 5, 6. The sifted key is then

$s = 01001$

Once the sifted key is created, Alice and Bob need to check it for errors. Errors can occur simply from the environment—causing bit flip and phase flip errors for example—or they can occur because an eavesdropper has tapped the channel.

If the error rate is too high, this probably indicates Eve is listening in. In this case, Alice and Bob check the error rate, and if it's above an agreed-upon threshold, they discard the key and start over.

The BB84 QKD protocol

- 1: Alice chooses $(4 + \delta)n$ random data bits.
- 2: Alice chooses a random $(4 + \delta)n$ -bit string b . She encodes each data bit as $\{|0\rangle, |1\rangle\}$ if the corresponding bit of b is 0 or $\{|+\rangle, |-\rangle\}$ if b is 1.
- 3: Alice sends the resulting state to Bob.
- 4: Bob receives the $(4 + \delta)n$ qubits, announces this fact, and measures each qubit in the X or Z basis at random.
- 5: Alice announces b .
- 6: Alice and Bob discard any bits where Bob measured a different basis than Alice prepared. With high probability, there are at least $2n$ bits left (if not, abort the protocol). They keep $2n$ bits.
- 7: Alice selects a subset of n bits that will serve as a check on Eve's interference, and tells Bob which bits she selected.
- 8: Alice and Bob announce and compare the values of the n check bits. If more than an acceptable number disagree, they abort the protocol.
- 9: Alice and Bob perform information reconciliation and privacy amplification on the remaining n bits to obtain m shared key bits.

Traffic Eavesdropping

- Let's suppose that Eve had made a measurement on bit 8. There is a 50% chance she selected the computational basis to make her measurement and a 50% chance that she selected the $|\pm\rangle$ basis. Suppose that she selected the computational basis. Then

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

- When Eve makes her measurement, the qubit will be either in the $|0\rangle$ state or the $|1\rangle$ state. Let's say it's $|0\rangle$. In the $\{|\pm\rangle\}$ basis this can be written as:

$$|0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}$$

- Although Alice prepared the qubit in the $|-\rangle$ state, when Bob makes his measurement, there is only a 50% chance of seeing the correct result. If we include a large number of qubits in our string, Alice and Bob can use this kind of behavior to deduce the presence of Eve. In our example Alice knows she created the bit string

01001

- But Bob has the bit string 01000
- Noise on the quantum channel will also create errors. Quantum error correction can be used to fix up those qubits.

AN EXAMPLE ATTACK: THE CONTROLLED NOT ATTACK

Suppose that Alice has the state

$$|+_A\rangle = \frac{|0_A\rangle + |1_A\rangle}{\sqrt{2}}$$

Can Eve duplicate this state in any way? Let's suppose that Eve wanted to make a state that gave the same measurement result for both Alice and Eve. What Eve can do is start with the state $|0_E\rangle$ and create the product state

$$|+_A\rangle \otimes |0_E\rangle = \frac{|0_A\rangle|0_E\rangle + |1_A\rangle|0_E\rangle}{\sqrt{2}}$$

Now watch what happens if Eve applies a controlled NOT gate to the state, using Alice's qubit as the control bit and Eve's qubit as the target. The state becomes

$$\frac{|0_A\rangle|0_E\rangle + |1_A\rangle|0_E\rangle}{\sqrt{2}} \xrightarrow{CN} \frac{|0_A\rangle|0_E\rangle + |1_A\rangle|1_E\rangle}{\sqrt{2}}$$

If Alice measures 0, we've got 0 for Eve. If Alice gets 1, Eve has 1. What happens if the measurement is made in the $|\pm\rangle$ basis? Then we have

$$\begin{aligned} \frac{|0_A\rangle|0_E\rangle + |1_A\rangle|1_E\rangle}{\sqrt{2}} &= \frac{1}{\sqrt{2}} \left[\left(\frac{|+_A\rangle + |-_A\rangle}{\sqrt{2}} \right) \left(\frac{|+_E\rangle + |-_E\rangle}{\sqrt{2}} \right) \right. \\ &\quad \left. + \left(\frac{|+_A\rangle - |-_A\rangle}{\sqrt{2}} \right) \left(\frac{|+_E\rangle - |-_E\rangle}{\sqrt{2}} \right) \right] \end{aligned}$$

$$\begin{aligned}
\frac{|0_A\rangle|0_E\rangle + |1_A\rangle|1_E\rangle}{\sqrt{2}} &= \frac{1}{\sqrt{2}} \left[\left(\frac{|+_A\rangle + |-_A\rangle}{\sqrt{2}} \right) \left(\frac{|+_E\rangle + |-_E\rangle}{\sqrt{2}} \right) \right. \\
&\quad \left. + \left(\frac{|+_A\rangle - |-_A\rangle}{\sqrt{2}} \right) \left(\frac{|+_E\rangle - |-_E\rangle}{\sqrt{2}} \right) \right] \\
&= \frac{1}{\sqrt{2}} \left(\frac{1}{2} \right) [|+_A\rangle|+_E\rangle + |+_A\rangle|-_E\rangle + |-_A\rangle|+_E\rangle + |-_A\rangle|-_E\rangle \\
&\quad + |+_A\rangle|+_E\rangle - |+_A\rangle|-_E\rangle - |-_A\rangle|+_E\rangle + |-_A\rangle|-_E\rangle] \\
&= \frac{1}{\sqrt{2}} (|+_A\rangle|+_E\rangle + |-_A\rangle|-_E\rangle)
\end{aligned}$$

Interestingly the correlation between Alice and Eve has been maintained! Eve's qubit assumes the same value as Alice's qubit in both bases. Suppose instead that Alice has

$$|-_A\rangle = \frac{|0_A\rangle - |1_A\rangle}{\sqrt{2}}$$

Does the correlation still hold? You can show that if we apply the same procedure, we end up with the state

$$|\psi\rangle = \frac{|+_A\rangle|-_E\rangle + |-_A\rangle|+_E\rangle}{\sqrt{2}}$$

We see that Alice and Eve get opposite measurement results. But Eve doesn't know what state Alice had ahead of time—so her measurement results are meaningless. Eve cannot, in general, form a product state with $|0_E\rangle$ and apply a controlled NOT gate to find out what Alice has.

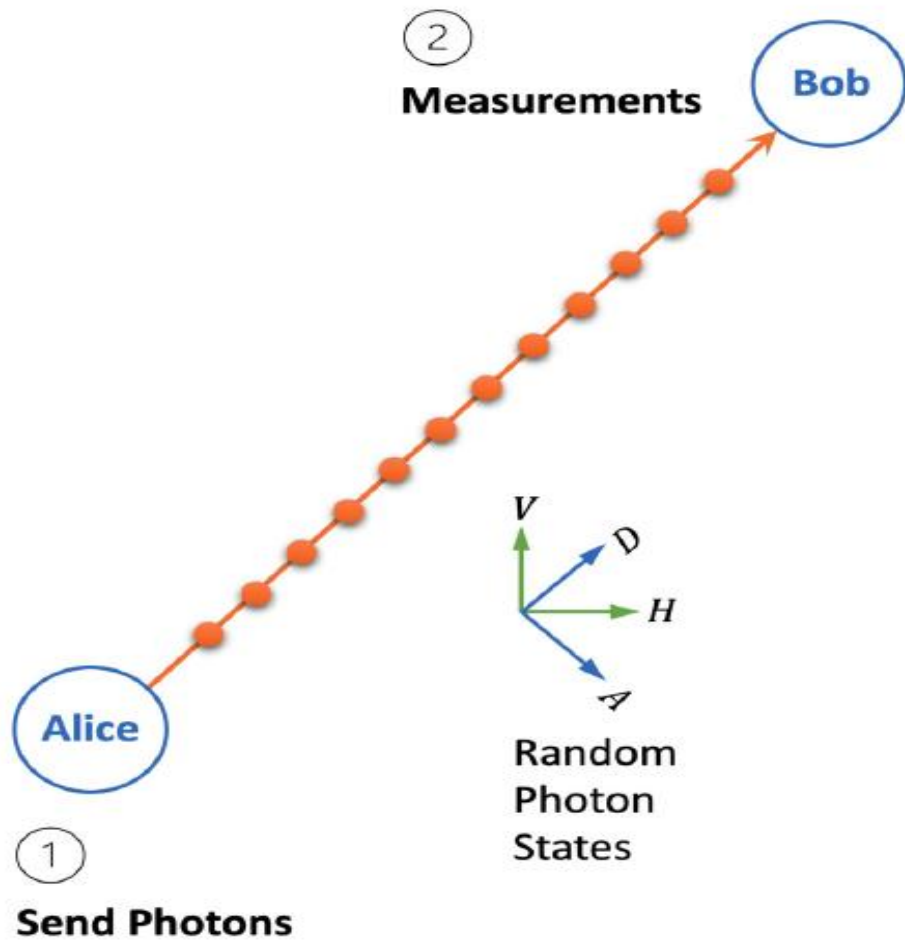
Representation of BB84 in different Books

Quantum Computing & Information: A Scaffolding Approach

Pg: 125

Quantum Computing for Computer Scientists book

Page: 268



Classical Channel

Id #	Alice Basis	Bob Basis	Basis Match	Alice's Results	Bob's Results	Eaves. Check
1	HV	DA		1	0	
2	DA	DA	✓	1	1	✓
3	HV	HV	✓	1	0	X
4	HV	DA		0	1	
5	DA	HV		1	1	
6	HV	HV	✓	0	0	✓
7	DA	DA	✓	0	1	X
8	HV	DA		1	0	
9	HV	HV	✓	0	0	
10	HV	HV	✓	1	1	
11	DA	DA	✓	1	1	
12	DA	HV		1	0	
13	HV	DA		0	1	
14	DA	HV		1	0	
15	DA	DA	✓	0	0	

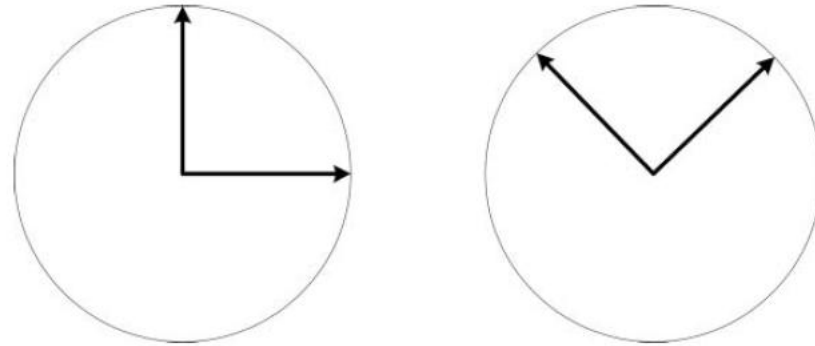
③ **Basis Announcement**

④ **Key Sifting**

⑤ **Eaves. Check**

rectilinear (HV) or
diagonal (DA) basis.

BB84



In this protocol, Alice will employ two different orthogonal bases

$$+ = \{ | \rightarrow \rangle, | \uparrow \rangle \} = \{ [1, 0]^T, [0, 1]^T \}$$

$$X = \{ | \nwarrow \rangle, | \nearrow \rangle \} = \left\{ \frac{1}{\sqrt{2}}[-1, 1]^T, \frac{1}{\sqrt{2}}[1, 1]^T \right\}.$$

We shall refer to the first basis as the “plus” basis and the second as the “times” basis. Essentially, they are two alternative vocabularies that Alice and Bob will use to communicate.

In these vocabularies, the states $|0\rangle$ and $|1\rangle$ shall be described by the following table:

State / Basis	$+$	$\times$
$ 0\rangle$	$ \rightarrow\rangle$	$ \nearrow\rangle$
$ 1\rangle$	$ \uparrow\rangle$	$ \nwarrow\rangle$

Step 1. Alice flips a coin n times to determine which classical bits to send. She then flips the coin another n times to determine in which of the two bases to send those bits. She then sends the bits in their appropriate basis.

For example, if n is 12, we might have something like this:

[illegible]

Step 2. As the sequence of qubits reaches Bob, he does not know which basis Alice used to send them, so to determine the basis by which to measure them he also tosses a coin n times. He then goes on to measure the qubit in those random bases.

In our example, we might have something like this:

Step 2: Bob receives n random bits in random measurements												
Bit number	1	2	3	4	5	6	7	8	9	10	11	12
Bob's random bases	X	+	X	X	+	X	+	+	X	X	X	+
Bob observes	↗	↑	↖	↖	↑	↗	↑	→	↖	↗	↖	→
Bob's bits	0	1	1	1	1	0	1	0	1	0	1	0

Step 3. Bob and Alice publicly compare which basis they used chose at each step. For instance, he can tell her X, +, X, X, Alice replies by telling him when he was right, and when he was not. Each time they disagreed, Alice and Bob scratch out the corresponding bit. Proceeding this way until the end, they are each left with a subsequence of bits that were sent and received in the same basis. If Eve was not listening to the quantum channel, this subsequence should be exactly identical. On average, this subsequence is of length $\frac{n}{2}$.

Step 3: Alice and Bob publicly compare bases used												
Bit number	1	2	3	4	5	6	7	8	9	10	11	12
Alice's random bases	+	+	X	+	+	+	X	+	X	X	X	+
Public channel	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕	⇕
Bob's random bases	X	+	X	X	+	X	+	+	X	X	X	+
Which agree?		✓	✓		✓			✓	✓	✓	✓	✓
Shared secret keys		1	1		1			0	1	0	1	0

Step 4. Bob randomly chooses half of the $\frac{n}{2}$ bits and publicly compares them with Alice. If they disagree by more than a tiny percentage (that could be attributed to noise), they know that Eve was listening in and then sending what she received. In that case, they scratch the whole sequence and try something else. If the exposed sequence is mostly similar, it means that either Eve has great guessing ability (improbable) or Eve was not listening in. In that case, they simply scratch out the revealed test subsequence and what remains is the unrevealed secret private key.

Step 4: Alice and Bob publicly compare half of the remaining bits												
Bit number	1	2	3	4	5	6	7	8	9	10	11	12
Shared secret keys		1	1		1			0	1	0	1	0
Randomly chosen to compare			✓						✓	✓		✓
Public channel			⇕						⇕	⇕		⇕
Shared secret keys		1	1		1			0	1	0	1	0
Which agree?			✓						✓	✓		✓
Unrevealed secret keys:		1			1			0			1	

B92 Protocol

- **B** stands for its inventor, Charles Bennett, and 1992 is the year it was published.

The main idea in B92 is that Alice uses only one *nonorthogonal* basis. Let us work out the protocol with the following example:

$$\{| \rightarrow \rangle, | \nearrow \rangle\} = \left\{ [1, 0]^T, \frac{1}{\sqrt{2}}[1, 1]^T \right\}.$$

$$+ = \{| \rightarrow \rangle, | \uparrow \rangle\} = \{[1, 0]^T, [0, 1]^T\}$$

$$X = \{| \nwarrow \rangle, | \nearrow \rangle\} = \left\{ \frac{1}{\sqrt{2}}[-1, 1]^T, \frac{1}{\sqrt{2}}[1, 1]^T \right\}.$$

Step 1. Alice flips a coin n times and transmits to Bob the n random bits in the appropriate polarization with a quantum channel.

Here is an example.

[illegible]

Step 2. For each of the n qubits, Bob measures the received qubits in either the $+$ basis or the X basis. He flips a coin to determine which basis to use.

There are several possible scenarios that can occur:

- If Bob uses the $+$ basis and observes a $|\uparrow\rangle$, then he knows that Alice must have sent a $|\nearrow\rangle = |1\rangle$ because if Alice had sent a $|\rightarrow\rangle$, Bob would have received a $|\rightarrow\rangle$.
- If Bob uses the $+$ basis and observes a $|\rightarrow\rangle$, then it is not clear to him which qubit Alice sent. She could have sent a $|\rightarrow\rangle$ but she could also have sent a $|\nearrow\rangle$ that collapsed to a $|\rightarrow\rangle$. Because Bob is in doubt, he will omit this bit.
- If Bob uses the X basis and observes a $|\nwarrow\rangle$, then he knows that Alice must have sent a $|\rightarrow\rangle = |0\rangle$ because if Alice had sent a $|\nearrow\rangle$, Bob would have received a $|\nearrow\rangle$.
- If Bob uses the X basis and observes a $|\nearrow\rangle$, then it is not clear to him which qubit Alice sent. She could have sent a $|\nearrow\rangle$ but she could also have sent a $|\rightarrow\rangle$ that collapsed to a $|\nearrow\rangle$. Because Bob is in doubt, he will omit this bit.

Step 2: Bob receives n random bits in a random basis

Bit number	1	2	3	4	5	6	7	8	9	10	11	12
Alice's random bits	→	→	↗	→	↗	→	↗	→	↗	↗	↗	→
Quantum channel	⇓	⇓	⇓	⇓	⇓	⇓	⇓	⇓	⇓	⇓	⇓	⇓
Bob's random bases	X	+	X	X	+	X	+	+	X	+	X	+
Bob's observations	↖	→	↗	↖	↑	↖	→	→	↗	↑	↗	→
Bob's bits	0	?	?	0	1	0	?	?	?	1	?	?

Step 3. Bob publicly tells Alice which bits were uncertain and they both omit them.

At this point, Alice and Bob know which bits are secret, so they may use those. But there is one more step if they want to detect whether or not Eve was listening in. They can, as in Step 4 of BB84, sacrifice half their hidden bits and publicly compare them. If they do not agree for a significant number, then they know that evil Eve has been doing her wicked deeds and the entire bit string should be ignored.

E91 Protocol (EPR Protocol)

- In 1991, Artur K. Ekert proposed a completely different type of quantum key distribution protocol based on entanglement (Ekert, 1991).

Step 1. Alice and Bob are each assigned one of each of the pairs of a sequence of entangled qubits.

When they are ready to communicate, they move on to Step 2.

Step 2. Alice and Bob separately choose a random sequence of bases to measure their particles. They then measure their qubits in their chosen basis.

An example might look like this:

Step 2: Alice and Bob measure in each of their random bases												
Bit number	1	2	3	4	5	6	7	8	9	10	11	12
Alice's random bases	X	X	+	+	X	+	X	+	+	X	+	X
Alice's observations	↗	↖	→	↑	↗	→	↖	→	→	↗	→	↗
Bob's random bases	X	+	+	X	X	+	+	+	+	X	X	+
Bob's observations	↗	→	→	↗	↗	→	↑	→	→	↗	↖	→

Step 3. Alice and Bob publicly compare what bases were used and keep only those bits that were measured in the same basis.

Step 3: Alice and Bob publicly compare their bases												
Bit number	1	2	3	4	5	6	7	8	9	10	11	12
Alice's random bases	X	X	+	+	X	+	X	+	+	X	+	X
Public channel	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
Bob's random bases	X	+	+	X	X	+	+	+	+	X	X	+
Which agree?	✓		✓		✓	✓		✓	✓	✓		

- If everything worked fine, Alice and Bob share a totally random secret key. But problems could have occurred.
- The entangled pairs could have been exposed to the environment and become disentangled or wicked Eve could have taken hold of one of the pairs, measured them, and sent along disentangled qubits.
- Alice or Bob randomly choose half of the remaining qubits and publicly compare the bits (not the bases). If they agree, then the last quarter of hidden qubits are probably good.
- If more than a small part disagree (from noise), then we must suspect Eve is up to her evil ways and our friendly communicators must throw away the entire sequence.